

Valid and ready

On-chip streaming interfaces move one beat per cycle when source and sink agree

Starter both assert

- Starter preset: Both assert, valid and ready both high on cycle zero, data hex A5
- Fire is one at cycle zero and exactly one beat transfers in the eight-cycle window
- Step through cycles and watch the verdict panel
- The fire row in the wave table marks every cycle where valid and ready overlap
- Toggle valid or ready cells to see fire disappear
- This is the same beat handshake you would use between a UART byte source and a FIFO sink

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Bus handshake animator

Step a conceptual valid/ready transfer: data moves only when both are high in the same cycle. Try source-stall and sink-stall patterns. Starter: both assert together → transfer on cycle 0.

Starter example: valid=ready=1 on cycle 0 with data 0xA5 — one beat transfers. Toggle cells or try stall presets.

Challenge 0 / 22 cleared

Quiz: transfer: In valid/ready handshake, a beat transfers when...

☐ valid and ready are both 1 in the same cycle

☐ only valid is 1

☐ only ready is 1

☐ the clock stops

Show hint

Check

Next

Load challenge setup

Idle

Quiz: transfer

Quiz: valid

Quiz: ready

Quiz: source stall

Quiz: sink stall

Quiz: hold

Quiz: where used

Quiz: combo ready

Transfer @@

One beat

Source stall xfer

Sink stall xfer

Three beats

Zero transfers

Handshake starter

Real RTL/TB practice

- In Track A, write the fire equation and explain valid and ready in one sentence each
- Sketch three cycles where valid is held high but ready is zero
- Optional: peek at streaming or AXI-Stream examples in the legacy materials and name the
- This lab is interface literacy, not a full protocol VIP or UART line driver

Pitfalls to watch

- Do not treat valid alone as a transfer, both must be one
- Do not assume the source drops valid every cycle; it may hold through a sink stall
- Real designs separate those paths
- Back-to-back bursts need ready high on consecutive cycles, not just one pulse
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, draw valid and ready for two cycles that fire and one that does not
- When you are ready, take the short quiz, then continue to self-check TB